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b) Simulate RIP using cisco packet tracer.

AIM:

To simulate RIP using cisco packet tracer.

Introduction:

Ip addres were configured on the PC and router interfaces. RIP was enabled on all Router interfaces. using the router rip command, and the connected networks were advertised using the network command. connectivity and route selection were verified using ping and traceroute.

Configure Router 0:

Router > enable.

Router # configure terminal.

Router (config)# interface fastethernet 0/1

Router (config-if) ip addres 10.0.0.1 255.0.0.0

Router (config-if) # no shutdown.

Router (config-if) # exit.

Router (config) # interface serial 0/0/0

Router (config-if)# ip address. 192.168.1.249.
255.255.255.252.

Router (config-if) # clock rate 64000

Router (config-if)# hardwidth 64

Router (config-if)# no shoutdown.

Router (config-if)# exit.

Router (config)# interface serial 0/0/1

Router (config-if) # ip addres 192.168.1.254
255.255.255.252

Router (config-if) # clock rate 64000

Router (config-if) # bandwidth 64.

Router (config-if) # no shutdown.

Router (config-if) # exit.

Configure Router 1:

Router > enable.

Router # configure terminal.

Router (config) # interface serial 0/0/0

Router (config-if) # ip address 192.168.1.245 255.255.255.252

Router (config-if) # no shutdown.

Router (config-if) # exit.

Router (config) # interface serial 0/0/1

Router (config-if) # ip address 192.168.1.254 255.255.255.252

Router (config-if) # clock rate 64000

Router (config-if) # no shutdown.

Router (config-if) # exit.

Configure Router 2.

Router > enable.

Router # configure terminal.

Router (config) # interface serial 0/0/0

Router (config-if) # ip address 192.168.1.245 255.255.255.252

Router (config-if) # no shutdown.

Router (config-if) # exit.

Router(config)# interface serial 0/0/1
Router(config-if)# ip address 192.168.1.253 255.255.255.252
Router(config-if)# no shutdown.
Router(config-if)# exit

Router(config)# interface fastEthernet 0/1
Router(config-if)# ip address 20.0.0.1 255.255.255.252.
Router(config-if)# no shutdown.
Router(config-if)# exit.

Configure RIP on Router0:

Router(config)# router rip.
255.252 Router(config-router)# version 2.
Router(config-router)# network 20.0.0.0
Router(config-router)# network 192.168.1.248.
Router(config-router)# network 192.168.1.252.
Router(config-router)# exit.

52 Configure RIP Router1:

Router(config)# router rip.
Router(config-router)# version 2.
Router(config-router)# network 192.168.1.248.
Router(config-router)# network 192.168.1.244.
Router(config-router)# exit

Configure RIP on Router2:

Router(config)# router rip
Router(config-router)# version 2.
Router(config-router)# network 20.0.0.0
Router(config-router)# network 192.168.1.252
Router(config-router)# network 192.168.1.244
Router(config-router)# exit

Verify RIP.

Router # show ip route.

Router # show ip protocals.

Test connectivity:

from PC0:

PC > ping 20.0.0.2.

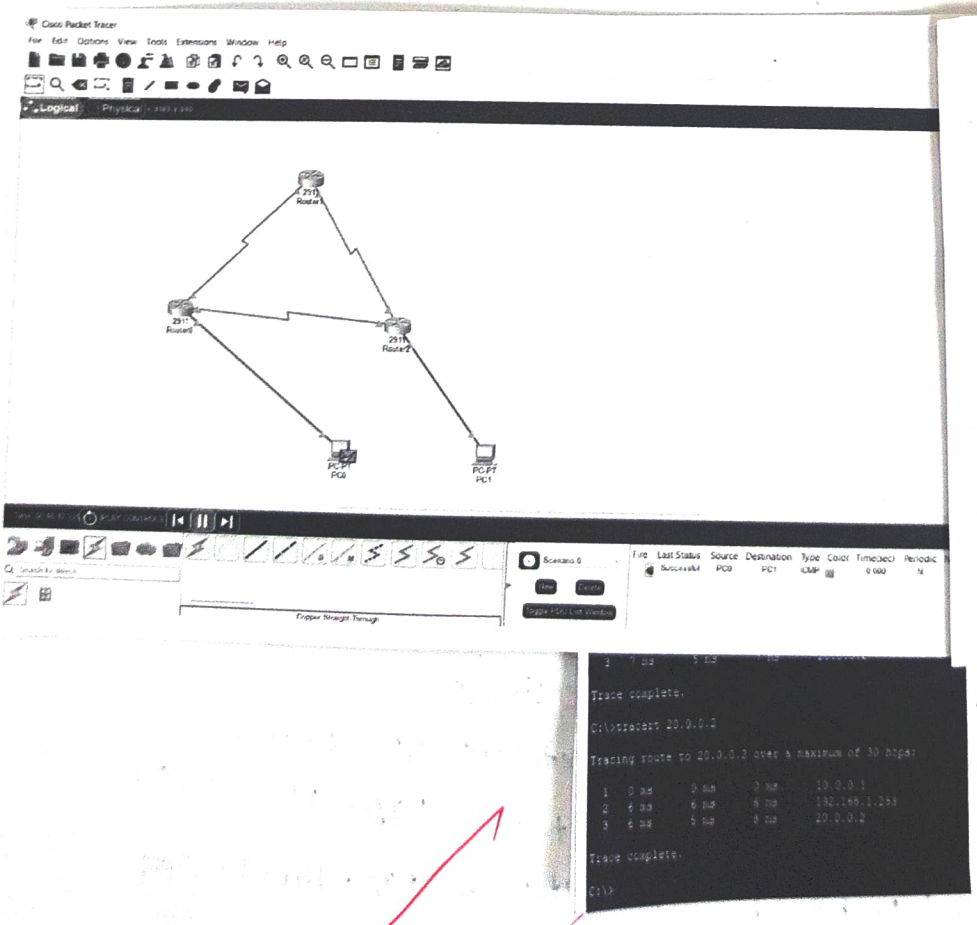
PC > tracert 20.0.0.2

Test alternative Route.

Disconnet the link between Router0 S0/0/1 and Router2 S0/0/1 then

PC > ping 20.0.0.2

PC > tracert 20.0.0.2



Result:

RIP was successfully configured and verified the network automatically selected the route with minum hop count and rerouted packets trough the alternate